Intro/ Description Paragraph -Review of Existing MIP Devices

To inform our design, we reviewed past sources that show different MIP devices and how they work. These devices use different mouth pieces, materials, display screens, and measurements. These devices also come with their own strengths and weaknesses including price, durability, and accuracy. Below is a summary and analysis of five different relevant devices:

1 Inspiratory Muscle Trainer Patent

- Description:
  - Patent for an inspiratory muscle trainer using a poppet valve and a plastic casing to maintain a pressure and require a certain amount of inspiratory pressure to open the valve allowing a patient to breathe
- Strengths:
  - It is very easy to adjust thanks to the poppet valve being replacable, it is very reliable so long as the spring doesn't fail, it is small enough to fit in a pocket, and it is very portable.
- Weaknesses:
  - The device is Fragile, complicated design might make it hard to replicate, spring requires occasional maintenance or else it might snap or bend.
- Relevance:
  - Well known patent for a relatively common IMT design. Elements such as the poppet valve can and should be brought into our design, small designs such as this are best, add on a removable mouthpiece and we could have a good design on our hands.
- Analysis
  - This design includes durability and portability because of its compact plastic casing, and adjustable valve. These all align with our key qualities of design by providing a light wright and semi-friendly tool. However, the case is very fragile and might not be optimal for long term use. Additionally, there is no digital display, so this hinders the user-friendly aspect of the design

2. MicroRPM Respiratory Pressure Meter

- **Description**:
  - The MicroRPM is a portable respiratory pressure meter used to measure MIP and MEP, designed for clinical use with advanced pressure-sensing technology and an optional PC software package for data analysis. *This design has been discontinued and replaced with the Pneumotrac RMS.*
- **Strengths**:
  - It offers high accuracy, portability, ease of use, compact design and optional data integration for enhanced functionality.
- **Weaknesses**:
  - Its high cost(*$2,200*), unnecessary features for education, and lack of easy cleaning make it impractical for large classroom settings. There is also a lack of battery life in this model.
- **Relevance**:
  - The MicroRPM demonstrates the importance of accuracy and portability in MIP devices while highlighting the need to simplify features and reduce costs for educational use.
- **Analysis**
  - Although the MicroRPM meets accuracy and usability standards, its high cost and maintenance challenges make it unsuitable for classroom settings. The design emphasizes portability and precision, which are valuable takeaways, but adapting these strengths into a simpler, affordable, and easy-to-clean device will better align with educational requirements.

3 Arduino MIP/MEP Device (Frontiers in Physiology)

- Description:
  - This MIP/MEP Arduino device is a low-cost, open-source tool designed to measure maximal inspiratory and expiratory pressures for respiratory muscle strength assessment.
- Strengths
  - Strengths include its affordability, which is €80 ($84.49 USD), ease of replication (parts are found mostly online) and adherence to established medical protocols for accurate and reliable pressure measurements (margin of error is ∓1%).

- Weaknesses
  - A potential weakness is its reliance on basic components (non-custom electronics), which may limit advanced features or customization in more specialized clinical environments.
- Relevance
  - This device's affordability and reliability make it ideal for Dr. Brossman’s lab, supporting student training and patient care in resource-constrained environments.
- Analysis
  - The Arduino device effectively demonstrates affordability, simplicity, and accuracy, fulfilling several key design priorities. Its use of inexpensive and widely available parts ensures replicability and accessibility in low-resource environments. While its basic components limit advanced features like digital displays, these are secondary priorities in our project. This model inspires our design by showcasing how a low-cost, functional tool can meet core needs while leaving room for improvements in durability and scalability.

4 Pressure threshold Training Device

- Description:
  - Pressure Threshold Training devices require a person to produce a sufficient amount of pressure to open the one way spring loaded valve contained within the device. The amount of pressure required to open the valve is adjusted by the knob on the device. The device will go silent when the patient is exhaling into the device.
- Strengths:
  - Calculates in a short period of time, with accurate and reliable results to support the patient's needs. The parts and materials are easily adjustable, with color coded measurement labels to easily identify the results of the device.
- Weaknesses:
  - This device's mouth pieces are very hard to clean/wash with a long drying period, with a high price (exact amount is not shown).

- Relevance:
    - This is relevant because it gives us a sense of what other MIP devices look like, and what Dr. Brossman wants and doesn't want.
- Analysis
    - The Pressure Threshold Training Devices excels in providing quick, reliable, and accurate results with adjustable pressure settings and intuitive labeling. The device also excels with simplicity and replicability. However, the devices impractical cleaning and drying process proves lackluster and fails to fulfill the durability/longevity requirement, making it unsuited for repeated use in a college classroom setting. The high cost fails the affordability requirement, and while generally accurate, the manual adjustments risk introducing errors, falling short of the accuracy standard. Additionally, the lack of protective storage diminishes portability and durability during constant transport, highlighting the need for improvements in these areas.

5 Vitalograph Pneumotrac Spirometer With RMS (MIP/MEP)

- Description
    - This Vitalograph Pneumotrac Spirometer with RMS measures maximal inspiratory pressure (MIP), maximal expiratory pressure (MEP), sniff nasal inspiratory pressure (SNIP), and peak cough flow (PCF).
    - Ideal for assessing respiratory muscle strength in patients with neuromuscular disease, pulmonary rehabilitation, and respiratory training response.
    - Spirometry features include pre- and post-bronchodilator testing, over 50 parameters, real-time feedback, and GOLD/GLI-based result interpretations.
    - Infection control with VBMax PFT Filters, automatic BTPS correction and USB power.
    - Key respiratory muscle strengths features include customizable reports, pressure/time curve display, and high resolution measurements (0.1 cm H_2O).
- Strengths
    - Comprehensive Assessment: Measures key respiratory metrics (MIP, MEP, SNIP, PCF) for thorough respiratory muscle evaluation

    - Advanced Software: Intuitive ATS/ERS-complaint Spirotrac 6 software with real time feedback, customization, and EHR/EMR compatibility
    - High precision and Infection Control: Offers 0.1 cm HS resolution and uses VBMax filters to prevent cross-contamination
    - Convenient and Reliable: USB-powered with a 5 year warranty, ideal for clinical settings
- Weaknesses
    - Very expensive. Since this is a multifunctional device, it costs more to produce, which isn't feasible for the project that we are completing.
    - Limited Portability: the USB powered and computer dependence might require a higher initial investment and user training
- Relevance
    - This gives us a good diagram of an MIP, helping us see what to and not include.
- Analysis
    - This multifunctional device is good for its advanced software features, making it highly accurate and reliable. While its USB-powered system and infection control measures are extremely important, the device's high cost and limited portability directly conflict with affordability and portability. Additionally, the complexity of its design contradicts the simplicity that we need for the students. Incorporating only the essential features from this design while reducing costs and simplifying the device would align with our goals.

<u>We identified key insights:</u>

1. Breath training effectiveness
2. Poppet valve
3. Portability
4. Affordability
5. Accuracy

<u>Explanation for each :</u>

1. **Breath Training Effectiveness:** One of the most important aspects of the device is its effectiveness in training the breathing of the user. If the device is unable to effectively measure the breathing, it defeats the purpose and will prevent the students/users from learning from it. All of the prior attempts of devices are efficient at breath training.
2. **Poppet Valve:** The Poppet Valve is a special sort of valve using a spring that only opens if enough air pressure is created allowing the device to not provide oxygen unless a certain amount of pressure is created. The Inspiratory Muscle Trainer Patent (IMT Patent) incorporates the poppet valve to allow adjustable pressures based on the diaphragmatic actions.
3. **Portability:** Since the device is being passed around by multiple students, it must be extremely portable. If it isn't, it poses a problem with the time constraint (with the students being unable to finish their analyses), transportation to various locations, etc. Both the Inspiratory Muscle Trainer and MicroRPM Respiratory Pressure Meter emphasize portability as a valuable feature, making devices easy to use in varied settings.
4. **Affordability:** Devices like the Arduino MIP/MEP offer an affordable solution compared to the costly MicroRPM and Vitalograph Pneumotrac, making them more accessible for classrooms with budget constraints.
5. **The accuracy:** The accuracy found in the MicroRPM and Vitalograph models sets a benchmark, showing that a high level of accuracy is critical for a credible MIP measurement tool. Ensuring that measurements are consistent is extremely important in a classroom setting.

Conclusion and Design Implications

The number one priority when designing the device should be making sure it actually provides a way to train the muscles of the lungs. A good design for the IMT should include any sort of way to reduce or stop air flow unless a certain amount of air pressure is created. Creating air pressure is the job of the diaphragm, and strengthening the diaphragm is the goal of an IMT device. A poppet valve is a good way to ensure that the device has this crucial quality. Portability is key to the success of our design, as the device must be able to be passed between students with ease. We must also make sure that our design is simple enough to be replicated at scale to make up to 50 devices. Certain aspects of some designs cannot be used because they require tuning and precision. The device will also need to be understandable to a student seeing the device for the first time. The device must also be durable enough to last for multiple years, fragile parts that would break easily cannot be used unless special care is taken. Precision and accuracy are also key when designing a device that does what we want it to do, if the device cannot accurately report the maximal inspiratory pressure of a user then the device is a failure.